

Dr. Thejus Mahajan
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Hamburg, 10 March 2026

Mrs. Dorothee Schaupp
Octapharma Biopharmaceuticals GmbH
Im Neuenheimer Feld 590
69120 Heidelberg

Application as Scientist R&D Molecular Design (m/f/d) – Job ID: 61464

Dear Mrs. Schaupp,

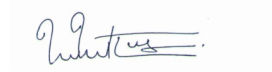
Optimizing therapeutic antibodies through phage display generates massive, high-dimensional NGS datasets. Mining this diversity and translating it into directed-evolution strategies requires robust, scalable bioinformatics workflows. As an experienced computational biologist, I am applying to serve as the dedicated computational bridge between your experimental screening teams and data-driven candidate optimization.

Over the past five years, I have specialized in translating complex biological questions into high-performance code. My core expertise aligns directly with your requirements:

- **NGS Pipelines (Nextflow/DSL2):** Built reproducible workflows that retrieve, curate, and process sequence data from biological databases (NCBI, MAFFT, trimAl). Ported computationally intensive Fortran models to a **Python/JAX** architecture enabling massive GPU/TPU parallelization—directly applicable to evaluating sequencing data for hit identification.
- **R & Biostatistics:** Refactored a monolithic R pipeline for clinical registry data (143K records, 2K variables) at HealthTwist GmbH, achieving 26% code reduction with byte-level reproducibility verification.
- **Cross-Functional Communication:** Fluent in the language of software engineering (Git, Linux, BASH) while possessing the biological understanding to translate protein engineering requirements into scripts, dashboards, and diversity analysis tools.

I am excited by the opportunity to contribute at Octapharma to the direct translation—the development of recombinant proteins with improved tolerability for patients. In a multidisciplinary unit, I see my strength in supporting experimental biology through superior algorithmic infrastructure and data analysis, and in driving the strategic evaluation of NGS data. I am available immediately and look forward to discussing how I can strengthen the bioinformatics workflows of your department in a personal interview.

Kind regards



Dr. Thejus Mahajan

Enclosures: Curriculum Vitae, Degrees and Certificates